Dual Wavelength Pulse Oximeter - Technical Datasheet

1. Device Description

A non-invasive medical tool called the Dual Wavelength Pulse Oximeter analyzes reflected light from two different wavelengths to determine blood oxygen saturation levels. Because of its low power consumption and portable form, it can be used for medical monitoring applications.

2. Applications

- Medical Monitoring Device
- Suitable for indoor use
- Not recommended near water or when wet
- Portable health monitoring system

3. Features

- Integrated LEDs, Photosensor, and 96x16 OLED screen
- Low power consumption
- Measures heart rate and blood oxygen levels

4. Electrical Characteristics (TA = 25°C)

Table 1: Electrical Characteristics

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Electrical Characteristics at TA=25ºC

Parameter	Device	Symbol	Min	Max	Units
Operating Voltage		V_O	3	3.3	V
Operating Current		I_O	600	1200	µA
Power Dissipation		P_O	1.8	3.96	mW
Operatin g		T_O	-40	+80	ºC
Tempera ture	Red LED	λ_R	635	635	nm
	Green LED	λ_G	568	568	
Peak Wavelen gth	Red LED	λ_R	625	635	nm
	Green LED	λ_G	570	670	
Dominant Wavelength					

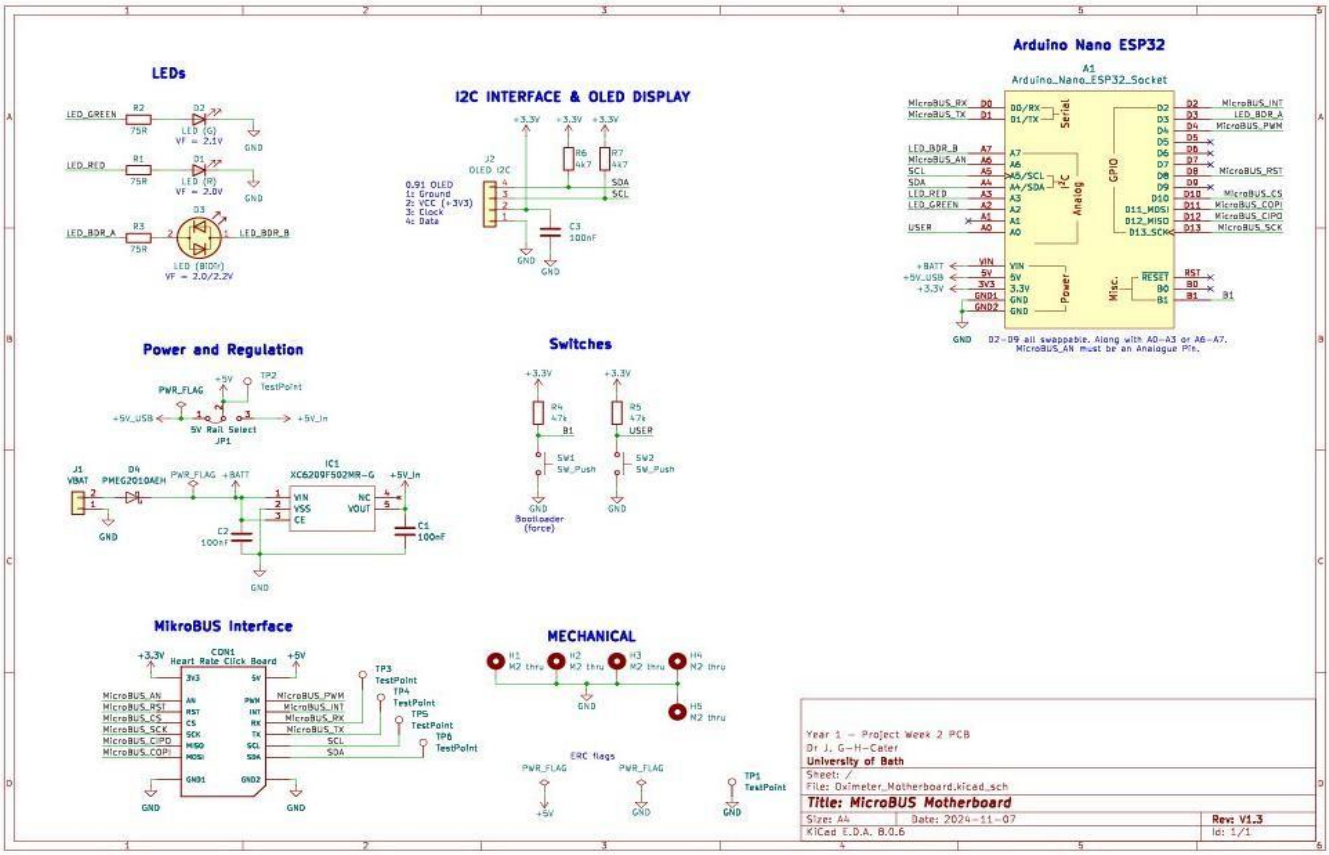


Figure 0: Physical view of the Dual Wavelength Pulse Oximeter device

5. Circuit & PCB Design

Figure 1: Schematic diagram of the Dual Wavelength Pulse Oximeter circuit and PCB design.

It includes the LED, photosensor, and microcontroller connections essential for accurate blood oxygen measurement.



6. Safety & Compliance

- Compliant with CE and IEC standards
- Contains lead in solder joints
- Components are repairable and reusable

7. Revision Control

- 06/01/2025 - First Release
- 07/01/2025 - Added Circuit & PCB Diagram and Table of Figures
- 08/01/2025 - Added Electrical Characteristics tables and safety compliance
- 09/01/2025 - Added Features and revised Electrical Characteristics

8. Performance Characteristics

The Dual Wavelength Pulse Oximeter is optimized for real-time, low-power monitoring. It operates at 3V to 3.3V with current draw between 600 µA and 1200 µA, enabling efficient energy use ideal for wearable applications. Its power dissipation ranges from 1.8 mW to 3.96 mW, and it functions reliably across temperatures from -40°C to +80°C. The integrated photosensor provides rapid response to changes in light absorption, ensuring accurate heart rate and oxygen saturation detection with minimal latency.

Contributors

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